

Q1 Khushi

a) Tautology

b) Modus Ponens - 1 mark

Truth Table - 1 mark

c) (2 marks)

$$CNF \rightarrow (\sim P \vee Q \vee R) \wedge$$

$$(\sim Q \vee P) \wedge (\sim R \vee P) \wedge (S \vee T)$$

2 marks

$$DNF \rightarrow (\sim P \wedge \sim Q \wedge \sim R \wedge S) \vee (\sim P \wedge \sim Q \wedge R \wedge T) \vee (P \wedge \sim Q \wedge S) \vee (P \wedge Q \wedge R \wedge T) \vee (R \wedge P \wedge S) \vee (P \wedge R \wedge T)$$

Q2 (Shreyas + Sanchit)

a) Calculation + Route 2

b) Calculation + Route 2

c) 6.84

d) Calculation + Route 3

e) UCB + Reason

f) i) $\{S_1, S_2, S_3\} \mid \{S_i \in \{\text{Normal, High}\}\}$

ii) Q learning + why q learning + why not others

iii) Backward induction + formula

iv) Apply backward induction for $T=2$

v) Yes + New State Space

Q3 (Eshika)

a) $\frac{3r}{2r+1}$

b) Correct proof Bayes conditional

c) i) 0.5

ii) $A = 99\%$ (no marks without calculation)
 $R = 99\%$
 $P = 50\%$

d) $k=1 \Rightarrow +1$ —

$k=3 \Rightarrow +1$ —

Graph —

Q4 Calculation of Gini
(YASH) Impurities ($1.5 \times 4 = 6$ Marks)

$$\text{Outlook} = 0.343$$

$$\text{Temperature} = 0.440$$

$$\text{Humidity} = 0.367$$

$$\text{Windy} = 0.429$$

Choosing min value as Outlook
(1 Mark)

Showing calculation
after Outlook split
- (3 Marks)

Final correct Decision Tree
- (2 Marks)

Q5 (9) Madhav, (1)
improves $\rightarrow$ accept
else acc with Prob < 1
(DE) $\checkmark$ (T) $\checkmark$ ($P = e^{\Delta E / T}$)
 $1/2$ $1/2$

(6) Pseudo ^{code} with clear
mention of variables
& conditions

①
 Prob < 1
 $e^{DE/T}$

h clear
 variables

Q6

RF1

Manav

1a) Bias increase

Variance decrease/
 remain same

RF2 $B \uparrow, =$
 $V \downarrow$

1b) $B \uparrow, V \downarrow$

For each part, 1 mark
 for correct answer and 1
 for reasoning.

Q7: SH

① Bias =

② Variance

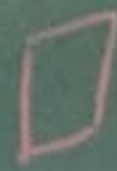
③ old m
 New

Partial

old = $0.25 +$

B_{part}

New = $12.25 +$



Q7: SHREYANSH

① Bias = $0.61 E[\hat{f}] - f$

② Variance = $E[(\hat{f} - E[\hat{f}])^2]$

③ Old MSE = 2.5
New " = 14.5

Partial marks for

old = $0.25 + 0 + \sigma^2$

Bias² Variance

New = $12.25 + 0 + \sigma^2$

Q8 (a) Proper heuristic function handling (YASH)

all row, col & box cases need value function not words - 2 marks
optimal value - 1 mark

Proof of how it achieves goal - 1 mark

(b) 4 states with proper heuristic value calculation (values needed for each state) - 3 Marks

Explaining neighbouring states calculation & choosing best state - 1 mark

(c) If No/Yes then explanation - 1 mark

If No → 1 counter example & calculation of values to prove it - 1 mark

If Yes → formal proof of how it is valid - 1 mark

Q9 ZAID

a- Nodes and edges (2 Points)

b- Order A, C, B, D

c- Order A, B, C, E or D

d- Not good as not consistent

e- provide values and argue